

Internal projectivity and properness of dualizable categories

Abstract

Relative internal projectivity implies properness for dualizable modules over a rigid $\mathbb{E}_1$ -monoidal category. We prove this by constructing a strongly continuous localization from a proper module and transferring a criterion for properness through compact morphisms in the internal Hom. The converse fails over every nonzero rigid base. Over a field, an explicit compactly generated presheaf category gives a counterexample: a diagram of Laurent modules admits no lift, even up to retract, through the induced functor on dualizable internal Homs. We describe the resulting integer inequalities and show that every countable restriction of this diagram does lift.

1 Introduction

Throughout this note, categories are ∞ -categories, and presentable categories are stable. We fix a universe; sets, indexing categories and colimits called small are taken in this universe. A functor is *continuous* if it preserves small colimits, and *strongly continuous* if it is continuous and its right adjoint is continuous. We write Fun^L and Fun^{LL} for the corresponding functor categories, Map for mapping spaces, and Hom for mapping spectra. Tensor products without a subscript are taken over Sp . Factorizations include a homotopy from the composite to the specified map; identities between functors or natural transformations are interpreted in their functor ∞ -categories.

Let $\mathcal{E}$ be a rigid presentable $\mathbb{E}_1$ -monoidal category in the sense of [EI, Definition 1.1]: its unit $1_{\mathcal{E}}$ is compact, and its multiplication admits a continuous $\mathcal{E}$ – $\mathcal{E}$ -linear right adjoint. The category $\mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$ has dualizable left $\mathcal{E}$ -modules as objects and strongly continuous $\mathcal{E}$ -linear functors as morphisms. Rigidity ensures that a continuous right adjoint to an $\mathcal{E}$ -linear functor is itself $\mathcal{E}$ -linear [EI, Proposition 1.12(iv)].

For $\mathcal{C} \in \mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$, its relative dual is the right $\mathcal{E}$ -module $\mathcal{C}^{\vee} = \mathrm{Fun}_{\mathcal{E}}^L(\mathcal{C}, \mathcal{E})$. The module $\mathcal{C}$ is *proper over $\mathcal{E}$* if evaluation

$$\mathrm{ev}_{\mathcal{C}/\mathcal{E}} : \mathcal{C} \otimes \mathcal{C}^{\vee} \longrightarrow \mathcal{E}$$

is strongly continuous. For $\mathcal{C}, \mathcal{D} \in \mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$, write $\underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{D})$ for their relative internal Hom in dualizable categories. It is an object of $\mathrm{Cat}_{\mathrm{st}}^{\mathrm{dual}}$, characterized by

$$\mathrm{Fun}^{LL}(\mathcal{A}, \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{D})) \simeq \mathrm{Fun}_{\mathcal{E}}^{LL}(\mathcal{A} \otimes \mathcal{C}, \mathcal{D}) \quad (1)$$

for dualizable $\mathcal{A}$; see [EI, Definition 3.1].

Universal evaluation induces a comparison functor

$$\rho_{\mathcal{D}} : \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{D}) \longrightarrow \mathrm{Fun}_{\mathcal{E}}^L(\mathcal{C}, \mathcal{D}).$$

For a strongly continuous $\mathcal{E}$ -linear functor $q : \mathcal{D} \rightarrow \mathcal{D}'$, its defining compatibility with evaluation gives a natural equivalence

$$\rho_{\mathcal{D}'} \circ \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, q) \simeq q_* \circ \rho_{\mathcal{D}}, \quad (2)$$

where q_* denotes postcomposition. Taking $\mathcal{A} = \mathrm{Sp}$ in (1) also gives

$$\underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{D})^{\omega} \simeq \mathrm{Fun}_{\mathcal{E}}^{LL}(\mathcal{C}, \mathcal{D}). \quad (3)$$

Here the left side denotes compact objects in the dualizable internal Hom; their ordinary comparison images are the functors on the right.

Definition 1.1. A dualizable left $\mathcal{E}$ -module $\mathcal{C}$ is *relatively internally projective* if

$$\underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, -) : \mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}} \longrightarrow \mathrm{Cat}_{\mathrm{st}}^{\mathrm{dual}}$$

takes short exact sequences to short exact sequences.

A short exact sequence consists of a fully faithful functor followed by its Verdier quotient, with both functors strongly continuous. Efimov proved relative internal projectivity for proper ω_1 -compact modules [EI, Theorem 3.6]. In that statement the relative diagonal, the image of the sphere under coevaluation in $\mathcal{C}^{\vee} \otimes_{\mathcal{E}} \mathcal{C}$, is required to be ω_1 -compact. For proper modules, this is equivalent to ω_1 -compactness as an object of $\mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$ [EI, Theorem 1.13]. He also suggested that internal projectivity should imply properness without a compactness assumption [EI, Section 3.3, before Proposition 3.19].

Theorem 1.2. *Let $\mathcal{E}$ be a rigid presentable stable $\mathbb{E}_1$ -monoidal category. Every relatively internally projective dualizable left $\mathcal{E}$ -module is proper over $\mathcal{E}$.*

Theorem 1.3. *For every nonzero rigid presentable stable $\mathbb{E}_1$ -monoidal category $\mathcal{E}$, there is a proper dualizable left $\mathcal{E}$ -module that is not relatively internally projective. When $\mathcal{E} = D(k)$ for a field k , such a module can be chosen compactly generated, with a set of compact generators whose pairwise mapping k -modules are either zero or k .*

The proof of Theorem 1.2 first expresses properness as a condition on relative Hom objects induced by compact morphisms (Proposition 2.1). We then construct a strongly continuous localization from a proper module to any dualizable module (Proposition 3.1). Internal projectivity allows the identity to be factored through compact morphisms above this localization. Section 4 uses that factorization to transfer the Hom criterion to the target.

For Theorem 1.3, Section 5 constructs a bipartite diagram of Laurent modules. A hypothetical lift in the internal Hom gives a compact morphism at each vertex, hence a bound on the Laurent denominators at that vertex. A diagonal choice among the target vertices contradicts one of these bounds. Section 6 carries out the same calculation using the compact unit of a rigid base. Section 7 gives a matrix criterion for lifting and explains why every countable restriction lifts although the full diagram does not.

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2 Compact morphisms and a criterion for properness

A morphism $f : x \rightarrow y$ in a presentable stable category is *compact* if, for every filtered diagram (z_i) and every map $y \rightarrow \mathrm{colim}_i z_i$, the composite from x factors through some z_i . Compact morphisms form a two-sided ideal. In a compactly generated category they are precisely the morphisms that factor through a compact object.

For objects x, y of a left $\mathcal{E}$ -module $\mathcal{A}$, write $\underline{\mathrm{Hom}}_{\mathcal{A}/\mathcal{E}}(x, y) \in \mathcal{E}$ for the relative internal Hom, characterized by

$$\mathrm{Map}_{\mathcal{E}}(e, \underline{\mathrm{Hom}}_{\mathcal{A}/\mathcal{E}}(x, y)) \simeq \mathrm{Map}_{\mathcal{A}}(e \otimes x, y).$$

Proposition 2.1. *A dualizable left $\mathcal{E}$ -module $\mathcal{A}$ is proper over $\mathcal{E}$ if and only if, for every pair of compact morphisms $f : x_0 \rightarrow x_1$ and $g : y_0 \rightarrow y_1$ in $\mathcal{A}$, the induced morphism*

$$\underline{\mathrm{Hom}}_{\mathcal{A}/\mathcal{E}}(x_1, y_0) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{A}/\mathcal{E}}(x_0, y_1), \quad h \longmapsto ghf, \quad (4)$$

is compact in $\mathcal{E}$.

We first describe the continuous approximations to the Hom functors that occur in this criterion. The needed factorization in the dual category is proved in Lemma 2.2.

2.1 Continuous Yoneda and dual objects

Let $\mathcal{A}$ be dualizable, and choose a sufficiently large uncountable regular cardinal κ , so that $\mathcal{E}$ and $\mathcal{A}$ are κ -presentable. Write $\mathcal{A}^\kappa$ for its full subcategory of κ -compact objects. The multiplication of $\mathcal{E}$ and its action on $\mathcal{A}$ are strongly continuous, by rigidity and [EI, Proposition 1.12(ii)]. Their continuous right adjoints imply that both functors preserve κ -compact objects. Thus $\mathcal{E}^\kappa$ is closed under tensor products and acts on $\mathcal{A}^\kappa$. The category $\mathrm{Ind}(\mathcal{A}^\kappa)$ carries the $\mathcal{E}$ -action obtained from its $\mathrm{Ind}(\mathcal{E}^\kappa)$ -action by restriction along the monoidal continuous Yoneda functor $\mathcal{E} \rightarrow \mathrm{Ind}(\mathcal{E}^\kappa)$ [EI, Proposition 1.16]. The continuous Yoneda functor and realization give an adjunction

$$\widehat{\mathcal{Y}} : \mathcal{A} \rightleftarrows \mathrm{Ind}(\mathcal{A}^\kappa) : \mathrm{colim}, \quad \widehat{\mathcal{Y}} \dashv \mathrm{colim}.$$

The functor $\widehat{\mathcal{Y}}$ is fully faithful and strongly continuous. Both functors are $\mathcal{E}$ -linear [EI, Propositions 1.16–1.17]. For $x \in \mathcal{A}^\kappa$, let $\mathcal{Y}x$ be its Yoneda image and let $c_x : \widehat{\mathcal{Y}}x \rightarrow \mathcal{Y}x$ be the counit at $\mathcal{Y}x$, using $\mathrm{colim} \mathcal{Y}x \simeq x$.

By [EL, Proposition 1.40], a morphism $f : x \rightarrow y$ between objects of $\mathcal{A}^\kappa$ is compact exactly when there is a lift

$$a_f : \mathcal{Y}x \longrightarrow \widehat{\mathcal{Y}}y, \quad c_y a_f \simeq \mathcal{Y}f. \quad (5)$$

The same criterion is available for arbitrary objects after increasing κ to contain them. A lift includes the displayed path in the mapping space. The triangle identities give $\mathrm{colim}(c_x) \simeq \mathrm{id}_x$, and hence $\mathrm{colim}(a_f) \simeq f$. Full faithfulness of $\widehat{\mathcal{Y}}$ then gives a path

$$a_f c_x \simeq \widehat{\mathcal{Y}}(f). \quad (6)$$

We will need to keep intermediate objects within a fixed $\mathcal{A}^\kappa$. This follows from the proof of [EL, Corollary 1.41], as follows. Given a compact $f : x \rightarrow y$ in $\mathcal{A}^\kappa$, write $\widehat{\mathcal{Y}}y \simeq \mathrm{colim}_i \mathcal{Y}z_i$ as a filtered colimit, with $z_i \in \mathcal{A}^\kappa$. Each structural map $z_i \rightarrow y$ is compact. Realization and continuity of $\widehat{\mathcal{Y}}$ give $\widehat{\mathcal{Y}}y \simeq \mathrm{colim}_i \widehat{\mathcal{Y}}z_i$. Since $\mathcal{Y}x$ is compact in $\mathrm{Ind}(\mathcal{A}^\kappa)$, the lift (5) factors through some $\widehat{\mathcal{Y}}z_i$. Consequently,

$$x \longrightarrow z_i \longrightarrow y \quad (7)$$

is a factorization into two compact morphisms in $\mathcal{A}^\kappa$.

For $x \in \mathcal{A}^\kappa$, define an object of $\mathcal{A}^\vee$ by

$$d_\kappa(x)(z) = \underline{\mathrm{Hom}}_{\mathrm{Ind}(\mathcal{A}^\kappa)/\mathcal{E}}(\mathcal{Y}x, \widehat{\mathcal{Y}}z). \quad (8)$$

Indeed, $\mathcal{Y}x$ is compact, hence relatively compact over $\mathcal{E}$ by [EI, Proposition 1.12(i)]. Thus (8) is continuous and $\mathcal{E}$ -linear in z . This defines a functor $d_\kappa : (\mathcal{A}^\kappa)^{\mathrm{op}} \rightarrow \mathcal{A}^\vee$. Applying realization to mapping objects gives a comparison

$$d_\kappa(x)(z) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{A}/\mathcal{E}}(x, z). \quad (9)$$

For a compact $f : x \rightarrow y$ in $\mathcal{A}^\kappa$, full faithfulness of $\widehat{\mathcal{Y}}$ and precomposition with a_f give a factorization

$$\underline{\mathrm{Hom}}_{\mathcal{A}/\mathcal{E}}(y, -) \longrightarrow d_\kappa(x) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{A}/\mathcal{E}}(x, -) \quad (10)$$

of precomposition with f . Its comparison with precomposition is a homotopy of natural transformations: apply the relative Hom functor $\underline{\mathrm{Hom}}_{\mathrm{Ind}(\mathcal{A}^\kappa)/\mathcal{E}}(-, \widehat{\mathcal{Y}}(-))$ to (6). Thus the factorization is natural in the target variable.

Lemma 2.2. *With κ as above, the following hold.*

- (i) *If $f : x \rightarrow y$ is compact in $\mathcal{A}^\kappa$, then $d_\kappa(f) : d_\kappa(y) \rightarrow d_\kappa(x)$ is compact in $\mathcal{A}^\vee$.*
- (ii) *Every compact morphism $u : \xi_0 \rightarrow \xi_1$ in $\mathcal{A}^\vee$ admits a factorization*

$$\xi_0 \longrightarrow d_\kappa(y_0) \xrightarrow{d_\kappa(h)} d_\kappa(y_1) \longrightarrow \xi_1,$$

where $h : y_1 \rightarrow y_0$ is compact and $y_0, y_1 \in \mathcal{A}^\kappa$.

Proof. The equivalence from relative to absolute duality is given by postcomposition with $\mathrm{Hom}_{\mathcal{E}}(1, -)$ [EI, Remark 1.18(iii)]. Under this equivalence, $d_\kappa(x)$ becomes the functor $z \mapsto \mathrm{Hom}_{\mathrm{Ind}(\mathcal{A}^\kappa)}(\mathcal{Y}x, \widehat{\mathcal{Y}}z)$. The equivalence commutes with precomposition and preserves compact morphisms. We may therefore work in the absolute dual categories.

Precomposition with realization and continuous Yoneda gives

$$\mathrm{colim}^* : \mathcal{A}^\vee \rightleftarrows \mathrm{Ind}((\mathcal{A}^\kappa)^{\mathrm{op}}) : \widehat{\mathcal{Y}}^*, \quad \mathrm{colim}^* \dashv \widehat{\mathcal{Y}}^*.$$

The left adjoint is fully faithful and strongly continuous. Denote by $\mathcal{Y}^\vee(y)$ the continuous extension of the compact covariant representable $\mathrm{Hom}_{\mathcal{A}}(y, -)|_{\mathcal{A}^\kappa}$. Then

$$\widehat{\mathcal{Y}}^* \mathcal{Y}^\vee(y) \simeq d_\kappa(y), \quad \mathrm{colim}^* d_\kappa(y) \longrightarrow \mathcal{Y}^\vee(y)$$

is the counit of this adjunction.

For (i), the lift a_f induces a factorization

$$\mathrm{colim}^* d_\kappa(y) \longrightarrow \mathcal{Y}^\vee(y) \longrightarrow \mathrm{colim}^* d_\kappa(x)$$

of $\mathrm{colim}^* d_\kappa(f)$. The middle object is compact. For the factorization homotopy, write ε for the counit of $\widehat{\mathcal{Y}} \dashv \mathrm{colim}$. For $w \in \mathrm{Ind}(\mathcal{A}^\kappa)$ and a map $k : \mathcal{Y}y \rightarrow \widehat{\mathcal{Y}} \mathrm{colim}(w)$, the composite is

$$\widehat{\mathcal{Y}} \mathrm{colim}(\varepsilon_w k) a_f \simeq k c_y a_f \simeq k \mathcal{Y}f.$$

These homotopies come from the counit's naturality and triangle identities and the chosen lift path, naturally in w ; they therefore give a path in the dual functor category. Since a fully faithful continuous functor reflects compact morphisms, $d_\kappa(f)$ is compact.

For (ii), factor u as two compact morphisms through an object ξ_m . After applying colim^* , each of these morphisms factors through a compact object. All compact objects of $\mathrm{Ind}((\mathcal{A}^\kappa)^{\mathrm{op}})$ are representable, because $\mathcal{A}^\kappa$ is stable and idempotent complete. We obtain

$$\mathrm{colim}^* \xi_0 \longrightarrow \mathcal{Y}^\vee(y_0) \longrightarrow \mathrm{colim}^* \xi_m \longrightarrow \mathcal{Y}^\vee(y_1) \longrightarrow \mathrm{colim}^* \xi_1.$$

By the adjunction, the map from $\mathrm{colim}^* \xi_m$ to $\mathcal{Y}^\vee(y_1)$ factors through $\mathrm{colim}^* d_\kappa(y_1)$. Thus the middle composite $\mathcal{Y}^\vee(y_0) \rightarrow \mathcal{Y}^\vee(y_1)$ has a lift to $\mathrm{colim}^* d_\kappa(y_1)$. Yoneda identifies this lift with a map $a_h : \mathcal{Y}y_1 \rightarrow \widehat{\mathcal{Y}}y_0$ lying over $\mathcal{Y}h$, where $h : y_1 \rightarrow y_0$ corresponds to that middle composite. The adjunction and Yoneda transport the triangle homotopy to $c_{y_0} a_h \simeq \mathcal{Y}h$. The criterion (5) shows that h is compact. Applying $\widehat{\mathcal{Y}}^*$ to the displayed factorization proves (ii). $\square$

2.2 Proof of the criterion

We use the compact-morphism test for relative evaluation from [EI, proof of Proposition 1.49(ii)]: $\text{ev}_{\mathcal{A}/\mathcal{E}}$ is strongly continuous if and only if $\text{ev}_{\mathcal{A}/\mathcal{E}}(f \boxtimes u)$ is compact for every compact morphism f in $\mathcal{A}$ and every compact morphism u in $\mathcal{A}^\vee$. The forward implication follows because external products of compact morphisms are compact and strongly continuous functors preserve compact morphisms.

Proof of Proposition 2.1. Suppose first that $\mathcal{A}$ is proper. For the given f, g , choose κ containing their four endpoints. Use (7) to factor $f \simeq f_2 f_1$ through $x_m \in \mathcal{A}^\kappa$, with both factors compact. By (10), the morphism (4) factors as

$$\begin{aligned} \underline{\text{Hom}}_{\mathcal{A}/\mathcal{E}}(x_1, y_0) &\longrightarrow d_\kappa(x_m)(y_0) \longrightarrow d_\kappa(x_0)(y_1) \\ &\longrightarrow \underline{\text{Hom}}_{\mathcal{A}/\mathcal{E}}(x_0, y_1). \end{aligned}$$

The middle arrow is evaluation on $g \boxtimes d_\kappa(f_1)$. It is compact by Lemma 2.2(i) and properness. The ideal property of compact morphisms proves the required conclusion.

Conversely, assume that (4) is always compact. It suffices to evaluate on $f \boxtimes u$, where $f : x_0 \rightarrow x_1$ is compact in $\mathcal{A}$ and u is compact in $\mathcal{A}^\vee$. Choose κ containing x_0, x_1 . By Lemma 2.2(ii) and the ideal property, it is enough to take $u = d_\kappa(h)$ for a compact $h : y_1 \rightarrow y_0$ in $\mathcal{A}^\kappa$. Factor $h \simeq h_2 h_1$ through $y_m \in \mathcal{A}^\kappa$, with both factors compact. Evaluation on $f \boxtimes d_\kappa(h)$ factors as

$$\begin{aligned} d_\kappa(y_0)(x_0) &\longrightarrow \underline{\text{Hom}}_{\mathcal{A}/\mathcal{E}}(y_0, x_0) \xrightarrow{h_2^*, f_*} \underline{\text{Hom}}_{\mathcal{A}/\mathcal{E}}(y_m, x_1) \\ &\longrightarrow d_\kappa(y_1)(x_1). \end{aligned} \tag{11}$$

Here the last arrow is induced by a lift $a_{h_1} : \mathcal{Y}y_1 \rightarrow \widehat{\mathcal{Y}}y_m$. To check the factorization, the precomposition map in $\text{Ind}(\mathcal{A}^\kappa)$ is

$$c_{y_0} \widehat{\mathcal{Y}}(h_2) a_{h_1} \simeq \mathcal{Y}h_2 c_{y_m} a_{h_1} \simeq \mathcal{Y}h.$$

The paths come from naturality of c , the chosen lift, and the factorization of h . Applying relative Hom and postcomposing with $\widehat{\mathcal{Y}}(f)$ identifies (11) with evaluation on $f \boxtimes d_\kappa(h)$ as maps in $\mathcal{E}$. The middle arrow of (11) is compact by assumption. Thus evaluation takes every compact external product to a compact morphism, and is strongly continuous. This proves properness. $\square$

3 Localizations from proper modules

Proposition 3.1. *For every $\mathcal{C} \in \text{Cat}_{\mathcal{E}}^{\text{dual}}$ there is a strongly continuous $\mathcal{E}$ -linear localization $q : \mathcal{B} \rightarrow \mathcal{C}$ such that $\mathcal{B}$ is proper over $\mathcal{E}$.*

To prove the proposition, we show that the class of modules admitting such a localization contains a colimit generator and is closed under coproducts and pushouts. The pushout construction requires a factorization of a continuous functor with proper source through a strongly continuous functor. Lemma 3.2 supplies the parameter category for that factorization.

We first record the closure properties of properness that will be used. Proper modules are closed under arbitrary small coproducts, including the empty coproduct, and under external

tensor product with a proper dualizable category over $\mathbf{Sp}$. They are also closed under strongly continuous fully faithful submodules. Indeed, if $i : \mathcal{A} \hookrightarrow \mathcal{B}$ is strongly continuous and fully faithful, then precomposition gives a strongly continuous fully faithful functor $(i^R)^* : \mathcal{A}^\vee \rightarrow \mathcal{B}^\vee$, with continuous right adjoint i^* , and

$$\mathrm{ev}_{\mathcal{A}/\mathcal{E}} \simeq \mathrm{ev}_{\mathcal{B}/\mathcal{E}} \circ (i \otimes (i^R)^*). \quad (12)$$

For a small set I , the coproduct $\mathcal{A} = \coprod_{i \in I} \mathcal{A}_i$ is represented by the product of the underlying presentable categories. A family of continuous functors $T_i : \mathcal{A}_i \rightarrow \mathcal{D}$ extends to $(a_i) \mapsto \coprod_i T_i(a_i)$, whose right adjoint is $d \mapsto (T_i^R(d))_i$. Colimits in a product category are computed coordinatewise, so this extension is strongly continuous whenever all T_i are. Restriction to the summands also identifies

$$\mathcal{A}^\vee \simeq \prod_i \mathcal{A}_i^\vee, \quad \mathcal{A} \otimes \mathcal{A}^\vee \simeq \prod_{i,j} (\mathcal{A}_i \otimes \mathcal{A}_j^\vee).$$

Evaluation on the (i, j) -summand is $\mathrm{ev}_{\mathcal{A}_i/\mathcal{E}}$ if $i = j$ and zero otherwise. Its right adjoint is therefore

$$(\mathrm{ev}_{\mathcal{A}/\mathcal{E}}^R(e))_{ij} \simeq \begin{cases} \mathrm{ev}_{\mathcal{A}_i/\mathcal{E}}^R(e), & i = j, \\ 0, & i \neq j. \end{cases}$$

Every coordinate preserves colimits when the $\mathcal{A}_i$ are proper, so the right adjoint is continuous for every small I . For external tensor products, evaluation is the tensor product of the two evaluations and is therefore strongly continuous.

We will also use the following generation criterion [EL, Proposition 1.8]. If a family of strongly continuous functors $i_s : \mathcal{A}_s \rightarrow \mathcal{B}$ has images generating $\mathcal{B}$ under colimits, then a continuous $F : \mathcal{B} \rightarrow \mathcal{D}$ is strongly continuous whenever every $F i_s$ is strongly continuous. To see this, the functors i_s^R are continuous and jointly conservative. They identify the colimit comparison for F^R with those for $(F i_s)^R$.

3.1 Proper parameter categories

Lemma 3.2. *Let $\mathcal{D}$ be a dualizable category over $\mathbf{Sp}$ and let $x \in \mathcal{D}$. There exist a proper dualizable category $\mathcal{P}$, a strongly continuous localization $p : \mathcal{P} \rightarrow \mathbf{Sp}$, and a strongly continuous functor $F : \mathcal{P} \rightarrow \mathcal{D}$ such that*

$$F(p^R \mathbb{S}) \simeq x.$$

Proof. Consider $\mathrm{Shv}_{\geq 0}(\mathbb{R}; \mathbf{Sp})$, the category of sheaves of spectra on $\mathbb{R}$ with singular support in $\mathbb{R} \times \mathbb{R}_{\geq 0}$. By [EL, Proposition 4.21(1)], it admits a strongly continuous fully faithful embedding into $\mathrm{Fun}(\mathbb{Q}_{\leq}^{\mathrm{op}}, \mathbf{Sp})$. The latter category is compactly generated by its representables, whose mutual mapping spectra are $\mathbb{S}$ or zero. Its evaluation therefore preserves compact generators of the tensor product and is strongly continuous. Formula (12) shows that $\mathrm{Shv}_{\geq 0}(\mathbb{R}; \mathbf{Sp})$ is proper.

There is a strongly continuous localization

$$p_{\mathbb{R}} : \mathrm{Shv}_{\geq 0}(\mathbb{R}; \mathbf{Sp}) \longrightarrow \mathbf{Sp}$$

whose right adjoint is the constant sheaf functor [EL, proof of Theorem 4.28, p. 83]. In particular, $p_{\mathbb{R}}^R \mathbb{S} \simeq \mathbb{S}_{\mathbb{R}}$. The description of strongly continuous functors from $\mathrm{Shv}_{\geq 0}(\mathbb{R}; \mathbf{Sp})$

by left-continuous rational diagrams, equivalently their left-continuous extensions to $\mathbb{R}$ [EL, Proposition C.4], together with [EL, Proposition D.3(1)], gives a Verdier quotient

$$\Phi : \text{Fun}^{LL}(\text{Shv}_{\geq 0}(\mathbb{R}; \text{Sp}), \mathcal{D}) \longrightarrow \mathcal{D}^{\omega_1}, \quad F_0 \longmapsto F_0(\mathbb{S}_{\mathbb{R}}).$$

Under the diagram description, Φ is the colimit functor. Each rational diagram has compact transition morphisms between distinct indices. By left continuity, its value at $a \in \mathbb{Q}$ is the colimit along a strictly increasing sequence of rational numbers tending to a from below. It is therefore ω_1 -compact by [EL, Propositions 1.24 and 1.40]. The diagram description identifies the source of Φ with a full subcategory of $\text{Fun}(\mathbb{Q}_{\leq}, \mathcal{D}^{\omega_1})$, so it is essentially small. It is stable and idempotent complete, since strongly continuous functors are closed under finite colimits and retracts, and Φ is exact. Choose a small skeleton before taking Ind . Consequently $\text{Ind}(\Phi)$ is a localization. It preserves compact objects and hence is strongly continuous.

Since $\mathcal{D}$ is ω_1 -compactly generated [EL, Corollary 1.21], choose an object $\tilde{x} \in \text{Ind}(\mathcal{D}^{\omega_1})$ whose realization is x . Express its image under the right adjoint as a small filtered Ind diagram:

$$\text{Ind}(\Phi)^R(\tilde{x}) \simeq \text{colim}_{i \in I} \mathcal{Y}(F_i), \quad F_i \in \text{Fun}^{LL}(\text{Shv}_{\geq 0}(\mathbb{R}; \text{Sp}), \mathcal{D}).$$

Applying $\text{Ind}(\Phi)$ and then realization gives a coherent diagram of strongly continuous functors such that

$$x \simeq \text{colim}_{i \in I} F_i(\mathbb{S}_{\mathbb{R}}). \quad (13)$$

Choose a small simplicial set K presenting I , and let $J = \Delta/K$ be its ordinary category of simplices. For two fixed simplices, the set of morphisms between them is a subset of the set of order-preserving maps between their finite indexing ordinals; in particular, it is finite. The last-vertex map $N(J) \rightarrow I$ is a localization [S, Theorem 1.3]. Its universal property identifies the spaces of cocones on a diagram on I with the spaces of cocones on its pullback to $N(J)$. Hence it preserves the colimit in (13). Applied to the constant point diagram, the same assertion gives $BJ \simeq BI \simeq *$, since I is filtered. Denote the pulled-back functors by F_j .

The category $\text{Fun}(J^{\text{op}}, \text{Sp})$ is proper: the mapping spectra between its compact representables are finite sums of sphere spectra. Moreover,

$$p_J = \text{colim}_{J^{\text{op}}} : \text{Fun}(J^{\text{op}}, \text{Sp}) \longrightarrow \text{Sp}$$

is a strongly continuous localization. Its constant right adjoint preserves colimits, and the counit is an equivalence because $BJ \simeq *$.

Set

$$\mathcal{P} = \text{Fun}(J^{\text{op}}, \text{Sp}) \otimes \text{Shv}_{\geq 0}(\mathbb{R}; \text{Sp}), \quad p = p_J \otimes p_{\mathbb{R}}.$$

The category $\mathcal{P}$ is proper and p is a strongly continuous localization. Under $\mathcal{P} \simeq \text{Fun}(J^{\text{op}}, \text{Shv}_{\geq 0}(\mathbb{R}; \text{Sp}))$, the diagram $j \mapsto F_j$ defines a continuous functor $F : \mathcal{P} \rightarrow \mathcal{D}$ by the presheaf universal property. Writing $y(j)$ for the spectral representable, it is characterized by $F(y(j) \boxtimes s) \simeq F_j(s)$. Its right adjoint is

$$F^R(d)(j) \simeq F_j^R(d).$$

For $j \rightarrow k$, the transition $F_k^R \rightarrow F_j^R$ is the right mate of $F_j \rightarrow F_k$. The mate correspondence therefore makes this a coherent J^{op} -diagram. Each F_j^R preserves colimits, so F^R does as well, by pointwise computation.

Finally, there is an equivalence $\text{const}_J \mathbb{S} \simeq \text{colim}_{j \in J} y(j)$: at an object k , the corresponding category of elements is k/J , which has the initial object id_k . It follows that

$$\begin{aligned} F(p^R \mathbb{S}) &\simeq F(\text{const}_J \mathbb{S} \boxtimes \mathbb{S}_{\mathbb{R}}) \\ &\simeq \text{colim}_{j \in J} F_j(\mathbb{S}_{\mathbb{R}}) \simeq x. \end{aligned}$$

□

3.2 Factorizations of continuous functors

Lemma 3.3. *Let $\mathcal{A}, \mathcal{B} \in \text{Cat}_{\mathcal{E}}^{\text{dual}}$, with $\mathcal{A}$ proper over $\mathcal{E}$. For every continuous $\mathcal{E}$ -linear functor $G : \mathcal{A} \rightarrow \mathcal{B}$, there are a proper left $\mathcal{E}$ -module $\mathcal{D}$, a strongly continuous localization $p_{\mathcal{A}} : \mathcal{D} \rightarrow \mathcal{A}$, and a strongly continuous $\mathcal{E}$ -linear functor $h : \mathcal{D} \rightarrow \mathcal{B}$ such that*

$$G \simeq hp_{\mathcal{A}}^R. \quad (14)$$

Proof. The category $\text{Fun}_{\mathcal{E}}^L(\mathcal{A}, \mathcal{B})$ is dualizable over Sp . One way to see this in the present setting is to use [EI, Proposition 3.4(i)]: the comparison

$$\underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{A}, \mathcal{B}) \longrightarrow \text{Fun}_{\mathcal{E}}^L(\mathcal{A}, \mathcal{B})$$

has a strongly continuous fully faithful left adjoint ℓ . It exhibits $\text{Fun}_{\mathcal{E}}^L(\mathcal{A}, \mathcal{B})$ as a continuous retract of a dualizable category. The same adjunction shows that ordinary evaluation is the composite

$$\text{Fun}_{\mathcal{E}}^L(\mathcal{A}, \mathcal{B}) \otimes \mathcal{A} \xrightarrow{\ell \otimes \text{id}_{\mathcal{A}}} \underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{A}, \mathcal{B}) \otimes \mathcal{A} \longrightarrow \mathcal{B}.$$

Thus this evaluation is strongly continuous.

Apply Lemma 3.2 to the object $G \in \text{Fun}_{\mathcal{E}}^L(\mathcal{A}, \mathcal{B})$. It gives $\mathcal{P}$, p , and a strongly continuous $F_0 : \mathcal{P} \rightarrow \text{Fun}_{\mathcal{E}}^L(\mathcal{A}, \mathcal{B})$ with $F_0(p^R \mathbb{S}) \simeq G$ in the functor category. Take

$$\mathcal{D} = \mathcal{P} \otimes \mathcal{A}, \quad p_{\mathcal{A}} = p \otimes \text{id}_{\mathcal{A}}, \quad h = \text{ev} \circ (F_0 \otimes \text{id}_{\mathcal{A}}).$$

The closure properties above show that $\mathcal{D}$ is proper. The functor $p_{\mathcal{A}}$ is a strongly continuous localization, and h is strongly continuous by the evaluation calculation. For $a \in \mathcal{A}$,

$$hp_{\mathcal{A}}^R(a) \simeq F_0(p^R \mathbb{S})(a) \simeq G(a).$$

This equivalence is obtained by evaluating the preceding equivalence of $\mathcal{E}$ -linear functors, and proves (14). □

3.3 Gluing and localization

For a continuous $\mathcal{E}$ -linear functor $h : \mathcal{A} \rightarrow \mathcal{B}$, define

$$\text{Gl}(h) = \mathcal{A} \times_{\mathcal{B}} \text{Fun}(\Delta^1, \mathcal{B}),$$

where the maps to $\mathcal{B}$ are h and target evaluation, and the pullback is taken in Cat_{∞} . Its objects are triples $(a, b, \phi : b \rightarrow h(a))$. The mapping space from (a, b, ϕ) to (a', b', ϕ') is

$$\text{Map}_{\mathcal{A}}(a, a') \times_{\text{Map}_{\mathcal{B}}(b, h(a'))}^h \text{Map}_{\mathcal{B}}(b, b'). \quad (15)$$

The maps to the base are $h(-)\phi$ and $\phi'(-)$; the homotopy pullback records the homotopy filling the square. If $\mathcal{A}$ and $\mathcal{B}$ are dualizable, then so is $\mathrm{Gl}(h)$ [EL, Proposition 1.78]. The two fully faithful inclusions

$$i_{\mathcal{A}}(a) = (a, 0), \quad i_{\mathcal{B}}(b) = (0, b)$$

are strongly continuous, with right adjoints

$$r_{\mathcal{A}}(a, b, \phi) = a, \quad r_{\mathcal{B}}(a, b, \phi) = \mathrm{fib}(b \rightarrow h(a)). \quad (16)$$

Their images generate $\mathrm{Gl}(h)$, as is seen from the natural cofiber sequence

$$i_{\mathcal{A}}(a) \longrightarrow (a, b, \phi) \longrightarrow i_{\mathcal{B}}(b) \longrightarrow i_{\mathcal{A}}(a)[1]. \quad (17)$$

This triangle also describes continuous functors from $\mathrm{Gl}(h)$. Given continuous $U : \mathcal{A} \rightarrow \mathcal{Z}$, $V : \mathcal{B} \rightarrow \mathcal{Z}$, and a natural transformation $\tau : Vh \rightarrow U[1]$, the corresponding functor is

$$(a, b, \phi) \longmapsto \mathrm{fib}(V(b) \xrightarrow{V(\phi)} Vh(a) \xrightarrow{\tau_a} U(a)[1]). \quad (18)$$

Finite limits in stable categories commute with colimits, so this functor is continuous. Conversely, applying a continuous functor to (17) recovers U, V, τ . The natural triangle and the fiber construction are inverse in the functor categories, including their natural transformations. The same argument applies to $\mathcal{E}$ -linear functors. This gives the functor description used below; compare [EL, Proposition 1.79].

Lemma 3.4. *If $\mathcal{A}$ and $\mathcal{B}$ are proper over $\mathcal{E}$ and h is strongly continuous, then $\mathrm{Gl}(h)$ is proper over $\mathcal{E}$.*

Proof. The functors $r_{\mathcal{A}}^*$ and $r_{\mathcal{B}}^*$ into $\mathrm{Gl}(h)^\vee$ are strongly continuous, with right adjoints $i_{\mathcal{A}}^*$ and $i_{\mathcal{B}}^*$. The latter are jointly conservative by (17); hence the images of $r_{\mathcal{A}}^*$ and $r_{\mathcal{B}}^*$ generate the dual. Evaluation on $i_X \otimes r_Y^*$, for $X, Y \in \{\mathcal{A}, \mathcal{B}\}$, is

$$(a, \xi) \longmapsto \mathrm{ev}_{Y/\mathcal{E}}(r_Y i_X(a), \xi).$$

The four functors $r_Y i_X$ are the identities, zero, and $h[-1]$, by (16). These restrictions of evaluation are therefore strongly continuous. The generation criterion proves the claim. $\square$

Lemma 3.5. *Suppose that $q_{\mathcal{A}} : \mathcal{A} \rightarrow \mathcal{A}'$ and $q_{\mathcal{B}} : \mathcal{B} \rightarrow \mathcal{B}'$ are strongly continuous $\mathcal{E}$ -linear localizations, and that $h : \mathcal{A} \rightarrow \mathcal{B}$ is continuous and $\mathcal{E}$ -linear. There is a strongly continuous localization*

$$\mathrm{Gl}(h) \longrightarrow \mathrm{Gl}(q_{\mathcal{B}} h q_{\mathcal{A}}^R). \quad (19)$$

Proof. The left adjoint takes (a, b, ϕ) to $(q_{\mathcal{A}} a, q_{\mathcal{B}} b)$, with structural map the composite

$$q_{\mathcal{B}} b \longrightarrow q_{\mathcal{B}} h(a) \longrightarrow q_{\mathcal{B}} h q_{\mathcal{A}}^R q_{\mathcal{A}} a.$$

For a target object (a', b', ϕ') , its right adjoint is given by

$$a = q_{\mathcal{A}}^R a', \quad b = q_{\mathcal{B}}^R b' \times_{q_{\mathcal{B}}^R q_{\mathcal{B}} h(a)} h(a), \quad (20)$$

with structural map the projection to $h(a)$. To check the adjunction, fix a source object (a_0, b_0, ϕ_0) . By (15), both mapping spaces in the claimed adjunction identify naturally with

$$\mathrm{Map}_{\mathcal{A}}(a_0, q_{\mathcal{A}}^R a') \times^h_{\mathrm{Map}_{\mathcal{B}}(b_0, q_{\mathcal{B}}^R q_{\mathcal{B}} h(q_{\mathcal{A}}^R a'))} \mathrm{Map}_{\mathcal{B}}(b_0, q_{\mathcal{B}}^R b').$$

On the source side this uses the pullback in (20); on the target side it uses the adjunctions for $q_{\mathcal{A}}$ and $q_{\mathcal{B}}$. All functors in this formula are continuous. Finite limits commute with colimits in stable categories, so the right adjoint is continuous. Applying $q_{\mathcal{A}}$ and $q_{\mathcal{B}}$ to the formula gives

$$q_{\mathcal{A}} a \simeq a', \quad q_{\mathcal{B}} b \simeq b' \times_{q_{\mathcal{B}} h(a)} q_{\mathcal{B}} h(a) \simeq b'.$$

The counit is thus an equivalence, proving the localization assertion. $\square$

3.4 Closure under pushouts

Proof of Proposition 3.1. Consider the full subcategory of $\mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$ consisting of modules that admit a strongly continuous localization from a proper module. It is closed under arbitrary small coproducts: for a family of such localizations $q_i : \mathcal{B}_i \rightarrow \mathcal{C}_i$, the right adjoint of $\coprod_i q_i$ is $(c_i) \mapsto (q_i^R(c_i))_i$. It is continuous and has an equivalence as counit, coordinatewise, while $\coprod_i \mathcal{B}_i$ is proper. We prove closure under pushouts.

Let

$$\mathcal{C}_1 \xleftarrow{F_1} \mathcal{C}_0 \xrightarrow{F_2} \mathcal{C}_2$$

be a span of strongly continuous functors, and choose strongly continuous localizations $q_i : \mathcal{B}_i \rightarrow \mathcal{C}_i$ with $\mathcal{B}_i$ proper. The functor

$$(q_1^R F_1 q_0, q_2^R F_2 q_0) : \mathcal{B}_0 \longrightarrow \mathcal{B}_1 \oplus \mathcal{B}_2$$

is continuous and $\mathcal{E}$ -linear. Lemma 3.3 gives a proper module $\mathcal{D}$, a strongly continuous localization $p : \mathcal{D} \rightarrow \mathcal{B}_0$, and a strongly continuous $h : \mathcal{D} \rightarrow \mathcal{B}_1 \oplus \mathcal{B}_2$ such that

$$hp^R \simeq (q_1^R F_1 q_0, q_2^R F_2 q_0).$$

By Lemma 3.4, $\mathrm{Gl}(h)$ is proper. Apply Lemma 3.5 to $q_0 p$ and $q_1 \oplus q_2$. Since

$$\begin{aligned} (q_1 \oplus q_2) h (q_0 p)^R &\simeq (q_1 \oplus q_2) h p^R q_0^R \\ &\simeq (F_1, F_2), \end{aligned}$$

we obtain a strongly continuous localization

$$\mathrm{Gl}(h) \longrightarrow \mathrm{Gl}((F_1, F_2) : \mathcal{C}_0 \rightarrow \mathcal{C}_1 \oplus \mathcal{C}_2). \quad (21)$$

The gluing on the right has three strongly continuous inclusions, denoted by i_0, i_1, i_2 . Invert the two natural transformations

$$i_1 F_1 \longrightarrow i_0[1], \quad i_2 F_2 \longrightarrow i_0[1]. \quad (22)$$

These are the connecting maps from (17). The resulting localization is the pushout $\mathcal{C}_1 \sqcup_{\mathcal{C}_0} \mathcal{C}_2$ in presentable categories. Indeed, by (18), a continuous functor from this gluing to a presentable stable category $\mathcal{Z}$ consists of continuous functors $H_i : \mathcal{C}_i \rightarrow \mathcal{Z}$ and transformations

$$H_1 F_1 \longrightarrow H_0[1], \quad H_2 F_2 \longrightarrow H_0[1].$$

Requiring both transformations to be equivalences leaves exactly H_1, H_2 , and an equivalence $H_1 F_1 \simeq H_2 F_2$, the data for a functor from the pushout. More precisely, its functor category is

$$\mathrm{Fun}^L(\mathcal{C}_1, \mathcal{Z}) \times_{\mathrm{Fun}^L(\mathcal{C}_0, \mathcal{Z})}^h \mathrm{Fun}^L(\mathcal{C}_2, \mathcal{Z}),$$

and passing to maximal subgroupoids gives the pushout mapping-space property. It suffices to impose (22) on a small set of generators, so this is an accessible localization.

This last localization is strongly continuous. Colimits of dualizable categories are computed in presentable categories, with strongly continuous structural functors [EL, Proposition 1.65]. The relative version follows by taking $\mathcal{E}$ -modules, using [EI, Proposition 1.12(v)]. Thus the restrictions of the localization to i_1 and i_2 are strongly continuous; its restriction to i_0 is a shifted composite of a structural functor with F_1 . The generation criterion applies. Composing this localization with (21) proves closure under pushouts.

Coproducts and pushouts generate all small colimits [HTT, Propositions 4.4.2.6–4.4.2.7]. Finally, $\mathcal{E} \otimes \mathrm{Shv}_{\geq 0}(\mathbb{R}; \mathrm{Sp})$ is proper over $\mathcal{E}$: the regular module $\mathcal{E}$ is proper by rigidity, and $\mathrm{Shv}_{\geq 0}(\mathbb{R}; \mathrm{Sp})$ is proper by the first paragraph of Lemma 3.2. This module generates $\mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$ under colimits [EI, Theorem 1.13]. The full subcategory under consideration is therefore all of $\mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$. $\square$

4 Relative internal projectivity

Lemma 4.1. *Let $Q : \mathcal{X} \rightarrow \mathcal{Z}$ be a strongly continuous localization between dualizable categories. For every compact $z \in \mathcal{Z}$, there are compact morphisms*

$$X_0 \xrightarrow{\alpha_1} X_1 \xrightarrow{\alpha_2} X_2$$

and maps $u : z \rightarrow QX_0$ and $v : QX_2 \rightarrow z$ such that

$$v Q(\alpha_2) Q(\alpha_1) u \simeq \mathrm{id}_z.$$

Proof. Realize a filtered Ind presentation of $\widehat{\mathcal{Y}}(Q^R z)$ as $Q^R z \simeq \mathrm{colim}_i x_i$. Each structural map $x_i \rightarrow Q^R z$ is compact by the continuous Yoneda criterion. Since Q is continuous and a localization, $z \simeq \mathrm{colim}_i Qx_i$. Compactness of z gives a factorization of id_z through some Qx_i , together with its path to the identity in $\mathrm{Map}_{\mathcal{Z}}(z, z)$. Factor $x_i \rightarrow Q^R z$ into two compact morphisms by [EL, Corollary 1.41]. Taking $X_0 = x_i$, $X_2 = Q^R z$, and v to be the counit proves the assertion. $\square$

Proof of Theorem 1.2. Let $\mathcal{C}$ be relatively internally projective. By Proposition 3.1, choose a strongly continuous localization $q : \mathcal{B} \rightarrow \mathcal{C}$ with $\mathcal{B}$ proper. Its kernel inclusion is strongly continuous: its right adjoint is given by $\mathrm{fib}(\mathrm{id}_{\mathcal{B}} \rightarrow q^R q)$, viewed in the kernel. This makes the kernel a continuous retract of $\mathcal{B}$, hence dualizable; rigidity gives the $\mathcal{E}$ -linearity of the right adjoint. Thus q belongs to a short exact sequence in $\mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$. Relative internal projectivity implies that

$$\underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, q) : \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{B}) \longrightarrow \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{C})$$

is a strongly continuous localization.

By (3), $\text{id}_{\mathcal{C}}$ corresponds to a compact object. Apply Lemma 4.1 to it, and then apply the comparison $\rho_{\mathcal{B}}$. Its naturality (2) gives continuous $\mathcal{E}$ -linear functors and natural transformations

$$F_0 \xrightarrow{\alpha_1} F_1 \xrightarrow{\alpha_2} F_2, \quad u : \text{id}_{\mathcal{C}} \longrightarrow qF_0, \quad v : qF_2 \longrightarrow \text{id}_{\mathcal{C}},$$

such that

$$v(q\alpha_2)(q\alpha_1)u \simeq \text{id}_{\text{id}_{\mathcal{C}}}. \quad (23)$$

This is a path between $\mathcal{E}$ -linear natural transformations in $\text{Fun}_{\mathcal{E}}^L(\mathcal{C}, \mathcal{C})$. The transformations α_1, α_2 here are images of compact morphisms in $\underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{C}, \mathcal{B})$.

Fix compact morphisms $f : a_0 \rightarrow a_1$ and $g : b_0 \rightarrow b_1$ in $\mathcal{C}$. Universal evaluation

$$\underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{C}, \mathcal{B}) \otimes \mathcal{C} \longrightarrow \mathcal{B}$$

is strongly continuous by (1). It sends the compact external products $\alpha_1 \boxtimes f$ and $\alpha_2 \boxtimes g$ to the compact morphisms

$$\begin{aligned} s &= \alpha_1(a_1)F_0(f) : F_0a_0 \longrightarrow F_1a_1, \\ t &= \alpha_2(b_1)F_1(g) : F_1b_0 \longrightarrow F_2b_1. \end{aligned}$$

Consider the following composite in $\mathcal{E}$:

$$\begin{aligned} \underline{\text{Hom}}_{\mathcal{C}/\mathcal{E}}(a_1, b_0) &\xrightarrow{F_1} \underline{\text{Hom}}_{\mathcal{B}/\mathcal{E}}(F_1a_1, F_1b_0) \\ &\xrightarrow{s^*, t_*} \underline{\text{Hom}}_{\mathcal{B}/\mathcal{E}}(F_0a_0, F_2b_1) \\ &\xrightarrow{q, u_{a_0}, v_{b_1}} \underline{\text{Hom}}_{\mathcal{C}/\mathcal{E}}(a_0, b_1). \end{aligned} \quad (24)$$

Since $\mathcal{B}$ is proper, Proposition 2.1 implies that the middle arrow is compact. The whole composite is therefore compact.

To identify this composite as a map in $\mathcal{E}$, put $M = \underline{\text{Hom}}_{\mathcal{C}/\mathcal{E}}(a_1, b_0)$ and let $e : M \otimes a_1 \rightarrow b_0$ be universal evaluation. The map $h \mapsto ghf$ is adjoint to

$$\beta = ge(\text{id}_M \otimes f) : M \otimes a_0 \longrightarrow b_1.$$

Under the same relative Hom adjunction, the composite (24) is represented by

$$\begin{aligned} M \otimes a_0 &\xrightarrow{\text{id}_M \otimes u_{a_0}} M \otimes qF_0a_0 \xrightarrow{\text{id}_M \otimes q(s)} M \otimes qF_1a_1 \\ &\xrightarrow{qF_1(e)} qF_1b_0 \xrightarrow{q(t)} qF_2b_1 \xrightarrow{v_{b_1}} b_1, \end{aligned}$$

where the arrow labeled $qF_1(e)$ uses the $\mathcal{E}$ -module structure of qF_1 . The definitions of s, t and module naturality of α_1, α_2, v identify this map with

$$\beta \circ (\text{id}_M \otimes (v_{a_0} q\alpha_2(a_0) q\alpha_1(a_0) u_{a_0})) \simeq \beta,$$

the last path being the component of (23). Transporting this path through the equivalence of mapping spaces defining relative Hom proves the required homotopy in $\mathcal{E}$. Thus (4) is compact for every f, g in $\mathcal{C}$. A final application of Proposition 2.1 proves that $\mathcal{C}$ is proper over $\mathcal{E}$. $\square$

Proposition 4.2 (Coproducts). *An arbitrary small coproduct of relatively internally projective dualizable left $\mathcal{E}$ -modules is relatively internally projective.*

Proof. Let $(\mathcal{C}_i)_{i \in I}$ be such a family. The universal property (1) and preservation of coproducts by tensor product give

$$\underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}\left(\coprod_{i \in I} \mathcal{C}_i, \mathcal{D}\right) \simeq \prod_{i \in I}^{\mathrm{dual}} \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}_i, \mathcal{D}).$$

The product here is taken in $\mathrm{Cat}_{\mathrm{st}}^{\mathrm{dual}}$. Such products preserve short exact sequences by [EL, Proposition 1.105]. Applying this result to the short exact sequences obtained from the individual $\mathcal{C}_i$ proves the assertion. The empty family gives the zero module. $\square$

5 A counterexample over a field

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. Define a small category J with objects

$$a_n \quad (n \in \mathbb{N}), \quad b_f \quad (f \in \mathbb{N}^{\mathbb{N}}),$$

and exactly one nonidentity arrow $u_{n,f} : a_n \rightarrow b_f$ for each pair (n, f) . There are no composable pairs of nonidentity arrows. For a field k , set

$$\mathcal{C}_k = \mathrm{Fun}(J^{\mathrm{op}}, D(k)), \quad p_v = k[J(-, v)].$$

Proposition 5.1. *The category $\mathcal{C}_k$ is compactly generated and proper over $D(k)$. For the localization*

$$q = - \otimes_{k[t]}^{\mathbf{L}} k[t, t^{-1}] : D(k[t]) \longrightarrow D(k[t, t^{-1}]),$$

the functor

$$\underline{\mathrm{Hom}}_{D(k)}^{\mathrm{dual}}(\mathcal{C}_k, q) : \underline{\mathrm{Hom}}_{D(k)}^{\mathrm{dual}}(\mathcal{C}_k, D(k[t])) \longrightarrow \underline{\mathrm{Hom}}_{D(k)}^{\mathrm{dual}}(\mathcal{C}_k, D(k[t, t^{-1}]))$$

has a compact object in its target which is not a retract of any object in its image. In particular, $\mathcal{C}_k$ is not relatively internally projective.

The mapping objects between representables will prove properness. We then construct a strongly continuous functor to $D(k[t, t^{-1}])$, hence a compact object of the target internal Hom . A hypothetical retract of an image object produces a natural transformation whose components are compact morphisms. Each component has a bounded denominator, while naturality at one of the vertices b_f forces arbitrarily large denominators.

5.1 Properness and the localization

The objects p_v form a set of compact generators of $\mathcal{C}_k$, and Yoneda gives

$$\underline{\mathrm{Hom}}_{\mathcal{C}_k/D(k)}(p_v, p_w) \simeq \begin{cases} k, & v = w, \\ k, & v = a_n, w = b_f, \\ 0, & \text{otherwise.} \end{cases} \quad (25)$$

The relative dual is compactly generated by the dual representables. Evaluation on an external product $p_v \boxtimes p_w^\vee$ is $\underline{\mathrm{Hom}}_{\mathcal{C}_k/D(k)}(p_w, p_v)$, hence is compact by (25). These external products generate $\mathcal{C}_k \otimes \mathcal{C}_k^\vee$, so evaluation preserves compact objects and has a continuous right adjoint. Thus $\mathcal{C}_k$ is proper.

The right adjoint Res of q is fully faithful and continuous. The kernel of q is the category $D_{t\text{-tors}}(k[t])$ generated by the compact object $k[t]/t$. The kernel inclusion has continuous right adjoint

$$M \longmapsto \mathrm{fib}(M \longrightarrow \mathrm{Res}(qM)).$$

Consequently,

$$0 \longrightarrow D_{t\text{-tors}}(k[t]) \longrightarrow D(k[t]) \xrightarrow{q} D(k[t, t^{-1}]) \longrightarrow 0 \quad (26)$$

is a short exact sequence in $\mathrm{Cat}_{D(k)}^{\mathrm{dual}}$.

The presheaf universal property defines a continuous $D(k)$ -linear functor $F : \mathcal{C}_k \rightarrow D(k[t, t^{-1}])$ by

$$F(p_v) = k[t, t^{-1}], \quad F(u_{n,f}) = t^{-f(n)}. \quad (27)$$

It sends every compact generator to a compact object, so it is strongly continuous. By (3), it corresponds to a compact object

$$\mathbf{F} \in \underline{\mathrm{Hom}}_{D(k)}^{\mathrm{dual}}(\mathcal{C}_k, D(k[t, t^{-1}]))^\omega.$$

5.2 Compact morphisms from a retract

We record the reduction from an arbitrary retract to compact morphisms. It will also apply over a general rigid base.

Lemma 5.2. *Let $T : \mathcal{X} \rightarrow \mathcal{Z}$ be a continuous functor, with $\mathcal{X}$ dualizable, and let $K \in \mathcal{Z}$ be compact. If K is a retract of $T(Y)$, there are a compact morphism $\beta : X \rightarrow Y$ and maps $u : K \rightarrow T(X)$, $e : T(Y) \rightarrow K$ such that*

$$eT(\beta)u \simeq \mathrm{id}_K.$$

Proof. Choose a sufficiently large regular cardinal κ and a filtered presentation $\widehat{\mathcal{Y}}(Y) \simeq \mathrm{colim}_i \mathcal{Y}(X_i)$ in $\mathrm{Ind}(\mathcal{X}^\kappa)$. Each structural map $X_i \rightarrow Y$ is compact: its Yoneda image factors through $\widehat{\mathcal{Y}}(Y) \rightarrow \mathcal{Y}(Y)$, so the assertion follows from [EL, Proposition 1.40]. Realization gives $Y \simeq \mathrm{colim}_i X_i$. Since T is continuous and K is compact, the retract section $K \rightarrow T(Y)$ factors through some $T(X_i)$. Take $\beta : X_i \rightarrow Y$ and retain the retract projection. $\square$

Lemma 5.3. *Let $\mathcal{E}$ be rigid, let $\mathcal{C}, \mathcal{D}, \mathcal{D}' \in \mathrm{Cat}_{\mathcal{E}}^{\mathrm{dual}}$, and let $q : \mathcal{D} \rightarrow \mathcal{D}'$ be a strongly continuous $\mathcal{E}$ -linear localization. Suppose that $F : \mathcal{C} \rightarrow \mathcal{D}'$ is strongly continuous and $\mathcal{E}$ -linear, and let $\mathbf{F} \in \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{D}')^\omega$ be its corresponding compact object. If $\mathbf{F}$ is a retract of an object in the image of $\underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, q)$, there are $X \in \underline{\mathrm{Hom}}_{\mathcal{E}}^{\mathrm{dual}}(\mathcal{C}, \mathcal{D})$ and a natural transformation*

$$\theta : \rho_{\mathcal{D}}(X) \longrightarrow q^R F$$

such that $\theta(c)$ is compact for every $c \in \mathcal{C}^\omega$, and $q\theta : q\rho_{\mathcal{D}}(X) \rightarrow F$ admits a natural section. Here the target of $q\theta$ is identified with F by the counit $qq^R F \simeq F$.

Proof. Lemma 5.2 gives a compact morphism $\beta : X \rightarrow Y$ in $\underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{C}, \mathcal{D})$ and maps

$$u : \mathbf{F} \longrightarrow \underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{C}, q)(X), \quad e : \underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{C}, q)(Y) \longrightarrow \mathbf{F}$$

whose composite with the image of β is the identity. By (2), the comparison image of e is a natural transformation $\bar{e} : q\rho_{\mathcal{D}}(Y) \rightarrow F$. Take its adjoint $\bar{e}^{\#} : \rho_{\mathcal{D}}(Y) \rightarrow q^R F$ and set $\theta = \bar{e}^{\#} \rho_{\mathcal{D}}(\beta)$. The adjunction identities give

$$q\theta \simeq \bar{e} q\rho_{\mathcal{D}}(\beta).$$

The comparison image of u is therefore a natural section.

For $c \in \mathcal{C}^{\omega}$, evaluation at c is the composite of universal evaluation with the tensor product of the identity and the strongly continuous functor $\text{Sp} \rightarrow \mathcal{C}$ classified by c . It is strongly continuous by [EI, Definition 3.1]. Strongly continuous functors preserve compact morphisms: apply their continuous right adjoints to a filtered test diagram and then use adjunction; see also [EL, proof of Proposition 1.46]. Thus $\rho_{\mathcal{D}}(\beta)(c)$ is compact, and so is its composite $\theta(c)$, since compact morphisms form a two-sided ideal. $\square$

5.3 The denominator contradiction

Proof of Proposition 5.1. Properness and compact generation were proved above. Suppose that $\mathbf{F}$ is a retract of an object in the image of $\underline{\text{Hom}}_{D(k)}^{\text{dual}}(\mathcal{C}_k, q)$. Apply Lemma 5.3 and write $X_v = \rho_{D(k[t])}(X)(p_v)$; for an arrow $u : v \rightarrow w$ of J , write $X(u) : X_v \rightarrow X_w$ for the map induced by the comparison functor. The lemma gives compact morphisms

$$\theta_v : X_v \longrightarrow \text{Res } k[t, t^{-1}]$$

whose localizations admit sections. Naturality gives

$$\theta_{b_f} X(u_{n,f}) \simeq t^{-f(n)} \theta_{a_n}. \quad (28)$$

These are homotopies of morphisms; after taking H^0 they become equalities of $k[t]$ -linear maps.

In $D(k[t])$ there is a filtered presentation

$$\text{Res } k[t, t^{-1}] \simeq \text{colim}_{N \geq 0} t^{-N} k[t].$$

Compactness of θ_v makes it factor through one stage. Thus, for some $N_v \geq 0$,

$$\text{im}(H^0 \theta_v) \subset t^{-N_v} k[t]. \quad (29)$$

Since $q\theta_{a_n}$ admits a section and localization is flat,

$$H^0(qX_{a_n}) = (H^0 X_{a_n})[t^{-1}] \longrightarrow k[t, t^{-1}]$$

is surjective. Choose a preimage of 1, written x_n/t^{d_n} with $x_n \in H^0 X_{a_n}$ and $d_n \geq 0$. In the Laurent polynomial ring,

$$(H^0 \theta_{a_n})(x_n) = t^{d_n}.$$

Define $f_*(n) = d_n + n$. The vertex b_{f_*} belongs to J , and (28) implies

$$(H^0 \theta_{b_{f_*}})(H^0 X(u_{n,f_*})(x_n)) = t^{-f_*(n)} t^{d_n} = t^{-n} \quad (n \geq 0).$$

This contradicts (29) at the single vertex b_{f_*} when $n > N_{b_{f_*}}$. Hence $\mathbf{F}$ is not a retract of any image object. The induced functor on internal Homs is not essentially surjective, so it cannot be a localization. Therefore $\underline{\text{Hom}}_{D(k)}^{\text{dual}}(\mathcal{C}_k, -)$ does not preserve the short exact sequence (26). $\square$

Remark 5.4. The lifting obstruction concerns the dualizable internal Hom. For the Laurent diagram F of Proposition 5.1, the ordinary continuous functor $\text{Res} \circ F$ always satisfies $q \text{ Res } F \simeq F$. Its values on the representable generators are the noncompact $k[t]$ -module $k[t, t^{-1}]$, so it is not strongly continuous. Properness gives the comparison functor a fully faithful strongly continuous left adjoint [EI, Proposition 3.4(i)]. Applying that left adjoint to $\text{Res } F$ gives an internal object whose localized ordinary comparison is F , by (2). Proposition 5.1 shows that the compact internal object classified by F is not a retract of this localized object. Thus equality of the ordinary comparison images does not identify the objects in the dualizable internal Hom.

6 The counterexample over every nonzero rigid base

Using the same category J , set

$$\mathcal{P} = \text{Fun}(J^{\text{op}}, \text{Sp}), \quad \mathcal{C}_{\mathcal{E}} = \mathcal{E} \otimes \mathcal{P} \simeq \text{Fun}(J^{\text{op}}, \mathcal{E}).$$

Proposition 6.1. *For every nonzero rigid presentable stable $\mathbb{E}_1$ -monoidal category $\mathcal{E}$, the module $\mathcal{C}_{\mathcal{E}}$ is proper over $\mathcal{E}$ and is not relatively internally projective.*

Properness follows by composing evaluation for $\mathcal{P}$ with multiplication in $\mathcal{E}$. To prove failure of internal projectivity, we use polynomial and Laurent polynomial actions in $\mathcal{E}$. Lemma 5.3 gives the same denominator bounds, and the compact unit detects the nonzero Laurent coefficients needed for the contradiction.

Proof. Properness. The spectral representables $p_v^{\text{Sp}} = \mathbb{S}[J(-, v)]$ compactly generate $\mathcal{P}$ and have mutual mapping spectra $\mathbb{S}$ or zero. As in (25), this proves that $\mathcal{P}$ is proper. The relative dual of $\mathcal{C}_{\mathcal{E}}$ is the right $\mathcal{E}$ -module $\mathcal{P}^{\vee} \otimes \mathcal{E}$, and relative evaluation is

$$\mathcal{E} \otimes \mathcal{P} \otimes \mathcal{P}^{\vee} \otimes \mathcal{E} \xrightarrow{\text{id} \otimes \text{ev}_{\mathcal{P}} \otimes \text{id}} \mathcal{E} \otimes \mathcal{E} \xrightarrow{\text{mult}_{\mathcal{E}}} \mathcal{E}.$$

Both arrows are strongly continuous; for the second this is part of rigidity [EI, Definition 1.1]. Thus $\mathcal{C}_{\mathcal{E}}$ is proper.

Polynomial localization. Let $\mathbb{S}[\mathbb{N}]$ and $\mathbb{S}[\mathbb{Z}]$ be the spectral monoid algebras. Their module categories in $\mathcal{E}$ consist of objects equipped, respectively, with an endomorphism and an automorphism. Inverting the endomorphism gives an $\mathcal{E}$ -linear localization

$$q_{\mathcal{E}} : \text{Mod}_{\mathbb{S}[\mathbb{N}]}(\mathcal{E}) \longrightarrow \text{Mod}_{\mathbb{S}[\mathbb{Z}]}(\mathcal{E}).$$

Here the algebra actions use the canonical action of spectra on $\mathcal{E}$. Its right adjoint Res is fully faithful and continuous. There are $\mathcal{E}$ -linear equivalences

$$\begin{aligned}\text{Mod}_{\mathbb{S}[\mathbb{N}]}(\mathcal{E}) &\simeq \mathcal{E} \otimes \text{Mod}_{\mathbb{S}[\mathbb{N}]}(\text{Sp}), \\ \text{Mod}_{\mathbb{S}[\mathbb{Z}]}(\mathcal{E}) &\simeq \mathcal{E} \otimes \text{Mod}_{\mathbb{S}[\mathbb{Z}]}(\text{Sp}).\end{aligned}$$

The spectral module categories are compactly generated, and rigidity makes $\mathcal{E}$ dualizable. These two module categories are therefore dualizable, also in the relative sense [EI, Proposition 1.12(iii)]. The kernel inclusion has continuous right adjoint $\text{fib}(\text{id} \rightarrow \text{Res } q_{\mathcal{E}})$, so the kernel of $q_{\mathcal{E}}$ is a continuous retract of the source and is dualizable [EL, Proposition 1.16]. Thus $q_{\mathcal{E}}$ occurs in a short exact sequence in $\text{Cat}_{\mathcal{E}}^{\text{dual}}$.

Write $1_{\mathcal{E}}[t]$ and $1_{\mathcal{E}}[t, t^{-1}]$ for the free polynomial and Laurent polynomial modules on $1_{\mathcal{E}}$. Both are compact, by the free-forgetful adjunction and compactness of $1_{\mathcal{E}}$. The equivalences

$$\text{Fun}_{\mathcal{E}}^L(\mathcal{E} \otimes \mathcal{P}, \mathcal{D}) \simeq \text{Fun}^L(\mathcal{P}, \mathcal{D}) \simeq \text{Fun}(J, \mathcal{D})$$

define a continuous $\mathcal{E}$ -linear functor $F_{\mathcal{E}} : \mathcal{C}_{\mathcal{E}} \rightarrow \text{Mod}_{\mathbb{S}[\mathbb{Z}]}(\mathcal{E})$ from the diagram with value $1_{\mathcal{E}}[t, t^{-1}]$ at every vertex and arrow maps $t^{-f(n)}$. In particular,

$$F_{\mathcal{E}}(e \otimes p_v^{\text{Sp}}) \simeq e \otimes 1_{\mathcal{E}}[t, t^{-1}].$$

Its right adjoint sends a Laurent module M to the presheaf with value the underlying object of M at every vertex and structure map $t^{-f(n)}$ along $u_{n,f}^{\text{op}}$. This follows from the free-forgetful adjunction for every $e \in \mathcal{E}$, and shows that the right adjoint preserves colimits. Hence $F_{\mathcal{E}}$ is strongly continuous and determines a compact object $\mathbf{F}_{\mathcal{E}}$ of the corresponding internal Hom by (3).

Laurent coefficients. The free representables $p_v^{\mathcal{E}} = 1_{\mathcal{E}} \otimes p_v^{\text{Sp}}$ are compact, since

$$\text{Map}_{\mathcal{C}_{\mathcal{E}}}(p_v^{\mathcal{E}}, Z) \simeq \text{Map}_{\mathcal{E}}(1_{\mathcal{E}}, Z(v)).$$

Suppose that $\mathbf{F}_{\mathcal{E}}$ is a retract of an object in the image of $\underline{\text{Hom}}_{\mathcal{E}}^{\text{dual}}(\mathcal{C}_{\mathcal{E}}, q_{\mathcal{E}})$. Let X be the object supplied by Lemma 5.3, and let X_v be its comparison value at $p_v^{\mathcal{E}}$, with $X(u_{n,f})$ the induced arrow map. Evaluation at these compact objects gives compact morphisms

$$\theta_v : X_v \longrightarrow \text{Res } 1_{\mathcal{E}}[t, t^{-1}]$$

whose localizations admit sections and which satisfy $\theta_{b_f} X(u_{n,f}) \simeq t^{-f(n)} \theta_{a_n}$. The presentation

$$\text{Res } 1_{\mathcal{E}}[t, t^{-1}] \simeq \text{colim}_{N \geq 0} t^{-N} 1_{\mathcal{E}}[t]$$

makes θ_v factor through $t^{-N_v} 1_{\mathcal{E}}[t]$ for some N_v . Here $t^{-N} 1_{\mathcal{E}}[t]$ is the free polynomial module with basis indexed by the degrees at least $-N$, mapping into the Laurent module by inclusion of those degrees.

Set $A = \pi_0 \text{Hom}_{\mathcal{E}}(1_{\mathcal{E}}, 1_{\mathcal{E}})$. Its identity 1_A is nonzero: otherwise $1_{\mathcal{E}}$ would be zero, forcing $\mathcal{E} = 0$. The free-forgetful adjunction and compactness of the unit give

$$\pi_0 \text{Map}_{\text{Mod}_{\mathbb{S}[\mathbb{N}]}(\mathcal{E})}(1_{\mathcal{E}}[t], \text{Res } 1_{\mathcal{E}}[t, t^{-1}]) \cong \bigoplus_{m \in \mathbb{Z}} A t^m = A[t, t^{-1}]. \quad (30)$$

Indeed, the underlying Laurent module is $\bigoplus_{m \in \mathbb{Z}} 1_{\mathcal{E}}$. The map from $t^{-N} 1_{\mathcal{E}}[t]$ gives precisely the summands in degrees at least $-N$. Hence postcomposition with θ_v induces a map

$$(\theta_v)_* : \pi_0 \operatorname{Map}_{\operatorname{Mod}_{\mathbb{S}[\mathbb{N}]}(\mathcal{E})}(1_{\mathcal{E}}[t], X_v) \longrightarrow A[t, t^{-1}]$$

whose image is contained in $t^{-N_v} A[t]$.

For a polynomial module Z , localization is the telescope of its t -action. The same adjunction and unit compactness therefore give a natural identification

$$\begin{aligned} \pi_0 \operatorname{Map}_{\operatorname{Mod}_{\mathbb{S}[\mathbb{Z}]}(\mathcal{E})}(1_{\mathcal{E}}[t, t^{-1}], q_{\mathcal{E}} Z) \\ \cong \left(\pi_0 \operatorname{Map}_{\operatorname{Mod}_{\mathbb{S}[\mathbb{N}]}(\mathcal{E})}(1_{\mathcal{E}}[t], Z) \right)[t^{-1}]. \end{aligned} \quad (31)$$

Here $(-)[t^{-1}]$ denotes the colimit of successive t -actions. Since $q_{\mathcal{E}} \theta_{a_n}$ admits a section, a preimage of 1_A in (31) can be written x_n/t^{d_n} , where

$$x_n \in \pi_0 \operatorname{Map}_{\operatorname{Mod}_{\mathbb{S}[\mathbb{N}]}(\mathcal{E})}(1_{\mathcal{E}}[t], X_{a_n}), \quad d_n \geq 0.$$

Consequently $(\theta_{a_n})_*(x_n) = t^{d_n} 1_A$. With $f_*(n) = d_n + n$, naturality implies that the image of $(\theta_{b_{f_*}})_*$ contains $t^{-n} 1_A$ for every $n \geq 0$. That image is contained in $t^{-N_{b_{f_*}}} A[t]$, contradicting the direct sum decomposition (30) and $1_A \neq 0$. Thus $\mathbf{F}_{\mathcal{E}}$ is not a retract of any image object. The functor $\underline{\operatorname{Hom}}_{\mathcal{E}}^{\operatorname{dual}}(\mathcal{C}_{\mathcal{E}}, q_{\mathcal{E}})$ is not a localization, and $\mathcal{C}_{\mathcal{E}}$ is not relatively internally projective. $\square$

Propositions 5.1 and 6.1 prove Theorem 1.3.

7 Lifting rank-one diagrams and the size condition

Fix a field k , and let $q : D(k[t]) \rightarrow D(k[t, t^{-1}])$ be localization at t . For small sets U, V , let $J_{U,V}$ have vertices a_u and b_v , with its only nonidentity arrows $a_u \rightarrow b_v$, one for each $(u, v) \in U \times V$. Empty index sets are allowed. Put

$$\mathcal{C}_{U,V} = \operatorname{Fun}(J_{U,V}^{\operatorname{op}}, D(k)), \quad Q_{U,V} = \underline{\operatorname{Hom}}_{D(k)}^{\operatorname{dual}}(\mathcal{C}_{U,V}, q).$$

For an integer matrix $m = (m_{uv})$, let $F_m : \mathcal{C}_{U,V} \rightarrow D(k[t, t^{-1}])$ be the continuous $D(k)$ -linear functor whose diagram on representables has value $k[t, t^{-1}]$ at every vertex and arrow $t^{-m_{uv}}$ from a_u to b_v . It is strongly continuous, since it takes the compact representable generators to compact objects. Write $\mathbf{F}_m$ for its corresponding compact object in $\underline{\operatorname{Hom}}_{D(k)}^{\operatorname{dual}}(\mathcal{C}_{U,V}, D(k[t, t^{-1}]))$.

Proposition 7.1. *The following conditions are equivalent:*

- (i) $\mathbf{F}_m$ belongs to the essential image of $Q_{U,V}$.
- (ii) $\mathbf{F}_m$ is a retract of an object in that essential image.
- (iii) There are integers λ_u and μ_v such that

$$m_{uv} \leq \lambda_u - \mu_v \quad (u \in U, v \in V). \quad (32)$$

When these conditions hold, F_m admits a lift whose value at each vertex is the free rank-one $k[t]$ -module.

Proof. Assume (ii). Lemma 5.3 gives a continuous $D(k)$ -linear functor $X : \mathcal{C}_{U,V} \rightarrow D(k[t])$ and a natural transformation $\theta : X \rightarrow \text{Res } F_m$. Its components θ_w at the representables are compact, and each $q\theta_w$ has a right inverse. Writing $X_w = X(p_w)$, the denominator argument of (29) shows that the $k[t]$ -submodules

$$L_w = \text{im}(H^0 X_w \longrightarrow k[t, t^{-1}])$$

are nonzero and satisfy $L_w \subset t^{-N_w} k[t]$ for some integers $N_w \geq 0$. For a nonzero Laurent polynomial h , let $\nu_t(h)$ be its least nonzero exponent. The values of ν_t on each $L_w \setminus \{0\}$ form a nonempty set of integers bounded below. Define

$$\lambda_u = \min_{0 \neq h \in L_{a_u}} \nu_t(h), \quad \mu_v = \min_{0 \neq h \in L_{b_v}} \nu_t(h).$$

Naturality gives $t^{-m_{uv}} L_{a_u} \subset L_{b_v}$. Applying this inclusion to an element realizing λ_u yields $\lambda_u - m_{uv} \geq \mu_v$, which is (iii).

Conversely, suppose (iii) holds. Take the diagram with value $k[t]$ at every vertex and arrow

$$t^{\lambda_u - \mu_v - m_{uv}} : k[t] \longrightarrow k[t]$$

from a_u to b_v . Its exponents are nonnegative, and its continuous extension $G : \mathcal{C}_{U,V} \rightarrow D(k[t])$ is strongly continuous because its values on the representable generators are compact. After localization, multiplication by t^{λ_u} at a_u and by t^{μ_v} at b_v defines a natural equivalence $qG \simeq F_m$:

$$t^{\mu_v} t^{\lambda_u - \mu_v - m_{uv}} = t^{-m_{uv}} t^{\lambda_u}.$$

The correspondence (1), with parameter Sp , therefore gives an actual lift of $\mathbf{F}_m$ under $\mathcal{Q}_{U,V}$. Finally, (i) implies (ii) by the identity retraction. $\square$

Proposition 7.2. *Every restriction of the diagram in Proposition 5.1 to a countable full subquiver admits a lift whose vertex values are free rank-one $k[t]$ -modules.*

Proof. First retain all source vertices a_n and a countable set of target vertices, indexed without repetition as b_{f_j} for $j \in I \subseteq \mathbb{N}$. Define

$$\begin{aligned} \lambda_n &= \max(\{0\} \cup \{f_j(n) : j \in I, j \leq n\}), \\ \mu_j &= \min(\{0\} \cup \{\lambda_n - f_j(n) : n < j\}). \end{aligned}$$

These extrema are taken over finite nonempty sets, including when I is finite or empty. If $n \geq j$, then $\lambda_n \geq f_j(n)$ and $\mu_j \leq 0$. If $n < j$, the definition of μ_j gives $\lambda_n - f_j(n) \geq \mu_j$. Thus $f_j(n) \leq \lambda_n - \mu_j$ for all n and $j \in I$. Proposition 7.1 supplies the lift. Restricting it further allows any subset of the source vertices. $\square$

Proposition 7.3. *The category $\mathcal{C}_k = \text{Fun}(J^{\text{op}}, D(k))$ of Proposition 5.1 is not ω_1 -compact in $\text{Cat}_{D(k)}^{\text{dual}}$.*

Proof. Let T range over the countable subsets of $\mathbb{N}^{\mathbb{N}}$, and let $J_T \subset J$ be the full subcategory on all a_n and the vertices b_f with $f \in T$. Set $\mathcal{C}_T = \text{Fun}(J_T^{\text{op}}, D(k))$. The poset of such T is ω_1 -filtered, since a countable union of countable sets is countable. For $i_T : J_T^{\text{op}} \hookrightarrow J^{\text{op}}$, left Kan extension

$$i_{T,!} : \mathcal{C}_T \longrightarrow \mathcal{C}_k$$

is fully faithful and strongly continuous: its right adjoint is restriction, which preserves colimits pointwise. The same holds for the transition functors between the $\mathcal{C}_T$.

The union of the J_T is J . Hence the limit of the categories $\mathcal{C}_T$ along restriction functors is $\mathcal{C}_k$. Passing to left adjoints computes the corresponding colimit in presentable $D(k)$ -linear categories. By [EL, Proposition 1.65] and [EI, Proposition 1.12(v)], it is also the colimit in $\text{Cat}_{D(k)}^{\text{dual}}$:

$$\mathcal{C}_k \simeq \operatorname{colim}_{T \subset \mathbb{N}^{\mathbb{N}}, |T| \leq \aleph_0} \mathcal{C}_T. \quad (33)$$

If $\mathcal{C}_k$ were ω_1 -compact, its identity would factor through some stage: there would be a strongly continuous $D(k)$ -linear $s : \mathcal{C}_k \rightarrow \mathcal{C}_T$ and a natural equivalence $i_{T,!} s \simeq \text{id}_{\mathcal{C}_k}$.

Choose $f \notin T$. Every object in the image of $i_{T,!}$ has value zero at b_f . Indeed, the comma category computing that value has objects given by arrows $w \rightarrow b_f$ in J^{op} with $w \in J_T$; there are no such arrows. Applying the asserted factorization to p_{b_f} and evaluating at b_f gives $0 \simeq k$, a contradiction. $\square$

For a proper module, ω_1 -compactness in $\text{Cat}_{D(k)}^{\text{dual}}$ is equivalent to ω_1 -compactness of its relative diagonal, by [EI, Theorem 1.13]; evaluation already preserves ω_1 -compact objects because it is strongly continuous. Thus the preceding proposition identifies precisely the compactness hypothesis absent from the field example. Its full diagram fails to lift although every countable restriction lifts.

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